Building a FEniCSx Ecosystem for Biomedical Research

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Translating physiological processes into predictive models requires robust and performant software pipelines. While DOLFINx offers a high-performance finite element backend, applying it to complex biomedical domains means bridging the gap between raw clinical data, specialized physical models, and scalable solvers. In this talk, we present a modular ecosystem of open-source FEniCSx-based tools designed to streamline workflows in brain and cardiac modelling.

For brain modelling, a major challenge is seamlessly integrating clinical imaging data into the finite element pipeline. We introduce MRI-toolkit, a specialized tool for converting MRI sequences into simulation-ready inputs, such as spatially varying concentration maps. We also discuss efficient strategies using wildmeshing and pyvista to generate high-quality computational meshes directly from medical image segmentations, bridging voxel data and FEniCSx-ready unstructured grids.

In cardiac modelling, we present a comprehensive suite of interoperable packages targeting various scales and physics. At the cellular level, gotranx facilitates the generation of ODE code for single-cell models. At the tissue and organ level, fenicsx-beat provides robust solvers for cardiac electrophysiology, while fenicsx-pulse simulates nonlinear cardiac mechanics. To support these models, we introduce cardiac-geometriesx for generating parametrised cardiac geometries, fenicsx-ldrb for assigning realistic myocardial fiber architectures, and circulation for coupling 3D finite element models to 0D lumped-parameter networks.

Finally, we share critical lessons learned from developing and maintaining this scientific software ecosystem, discussing our approaches to automated testing, package distribution (PyPI/Conda/Spack), and creating user-friendly documentation.

References

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